Locality-Attending Vision Transformer

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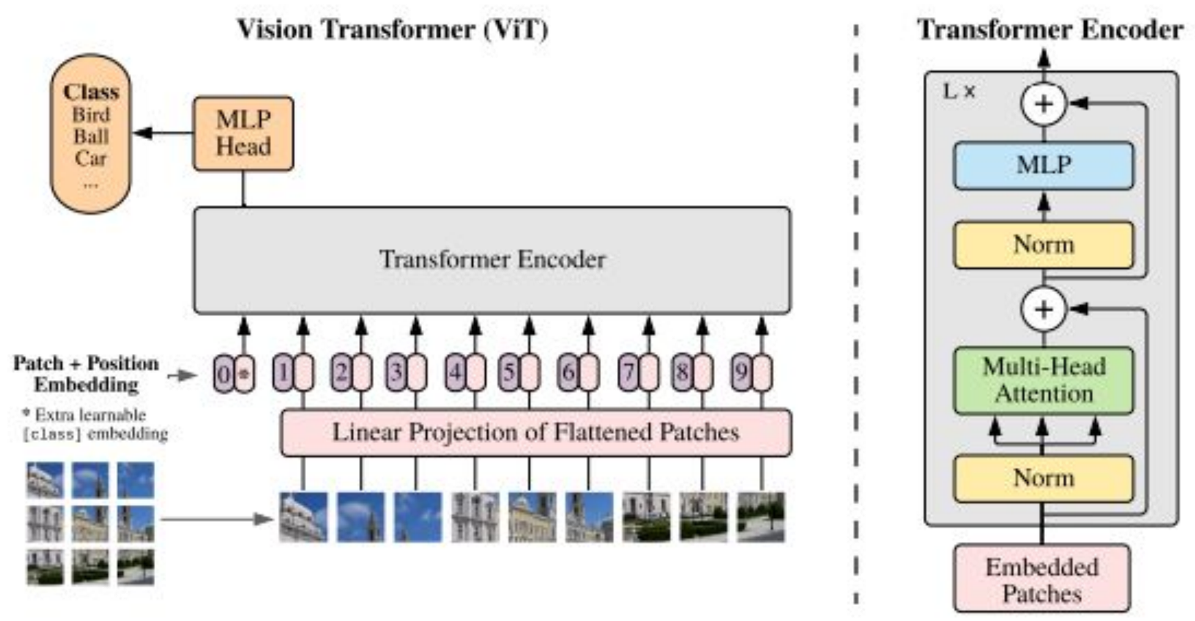
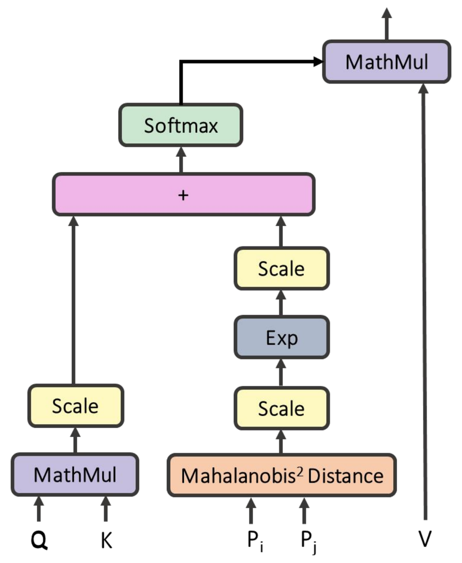
Motivation and Idea

Vanilla ViT: → Uses global self-attention, Excels at image classification, **Struggles at segmentation** (dense prediction tasks) → Why? → Global attention dilutes fine spatial details, Only the [CLS] token receives supervision during classification training.



Gaussian-Augmented Attention (GAug)

We modify the self-attention mechanism by adding a patch-specific, learnable Gaussian locality bias to the attention logits. This modification is applied to all transformer layers and to all spatial tokens (excluding the [CLS] token).



$$p_i = (x_i, y_i)$$

$$D_{ij} = (p_i - p_j)^T \Sigma^{-1} (p_i - p_j)$$

$$S_{ij} = \alpha \exp\left(-\frac{1}{2} D_{ij}\right)$$

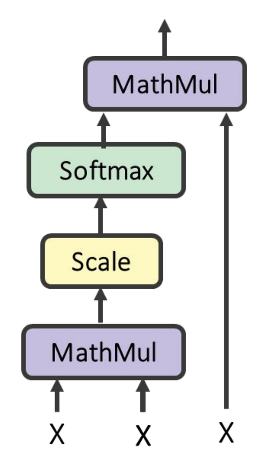
$$\tilde{L}_{ij} = \frac{q_i k_j^T}{\sqrt{d}} + S_{ij}$$

$$\tilde{A}_{ij} = \mathrm{softmax}_j(\tilde{L}_{ij})$$

$$\mathrm{GAug}(X)_i = \sum_{j=1}^{T} \tilde{A}_{ij} v_j$$

Patch Representation Refinement (PRR)

When training a ViT for classification, the loss is computed only from the CLS token. As a result, spatial patch features remain under-trained. To encourage meaningful patch representations we introduce PRR at the final layer's output, applying it after the last transformer block and before the classification head.



$$X \in \mathbb{R}^{T \times d}$$

$$M = \frac{XX^T}{\sqrt{d}}$$

$$R = \mathrm{softmax}_j(M)$$

$$R_{ij} = \frac{\exp(M_{ij})}{\sum_k \exp(M_{ik})}$$

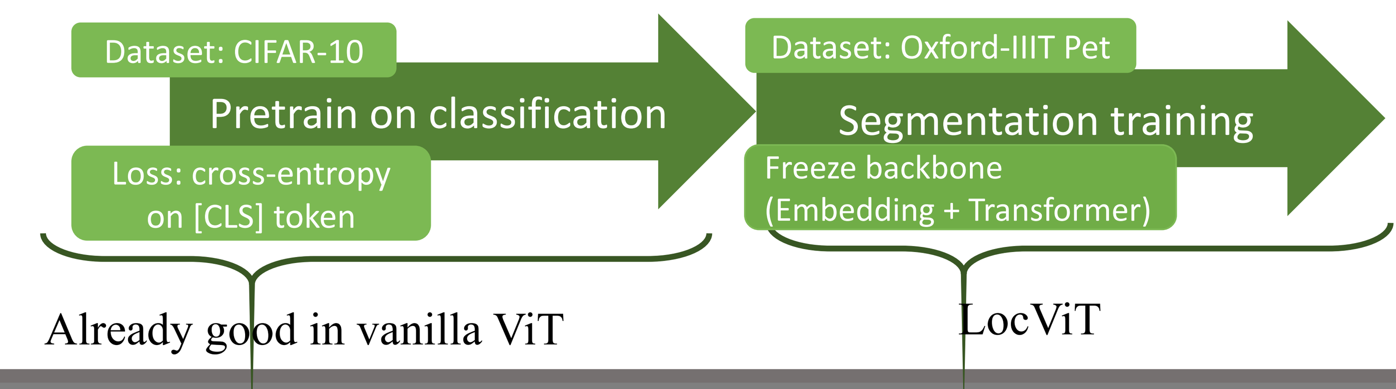
$$X^+ = RX$$

$$x_i^+ = \sum_{j=1}^{T} R_{ij} x_j$$

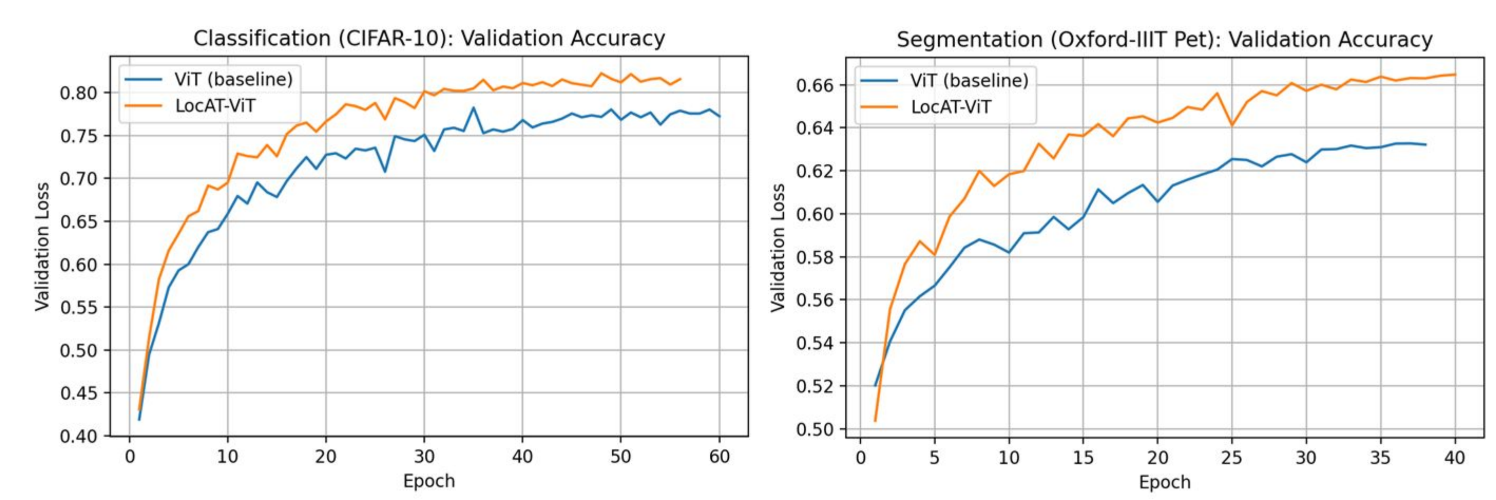
Training process

We introduce a modular add-on to improve spatial locality without modifying the classification objective: **LocAt = GAug + PRR**

During segmentation, the pretrained backbone is frozen and only a 1-layer segmentation head is trained to produce per-patch logits, which are then upsampled to the original image resolution.



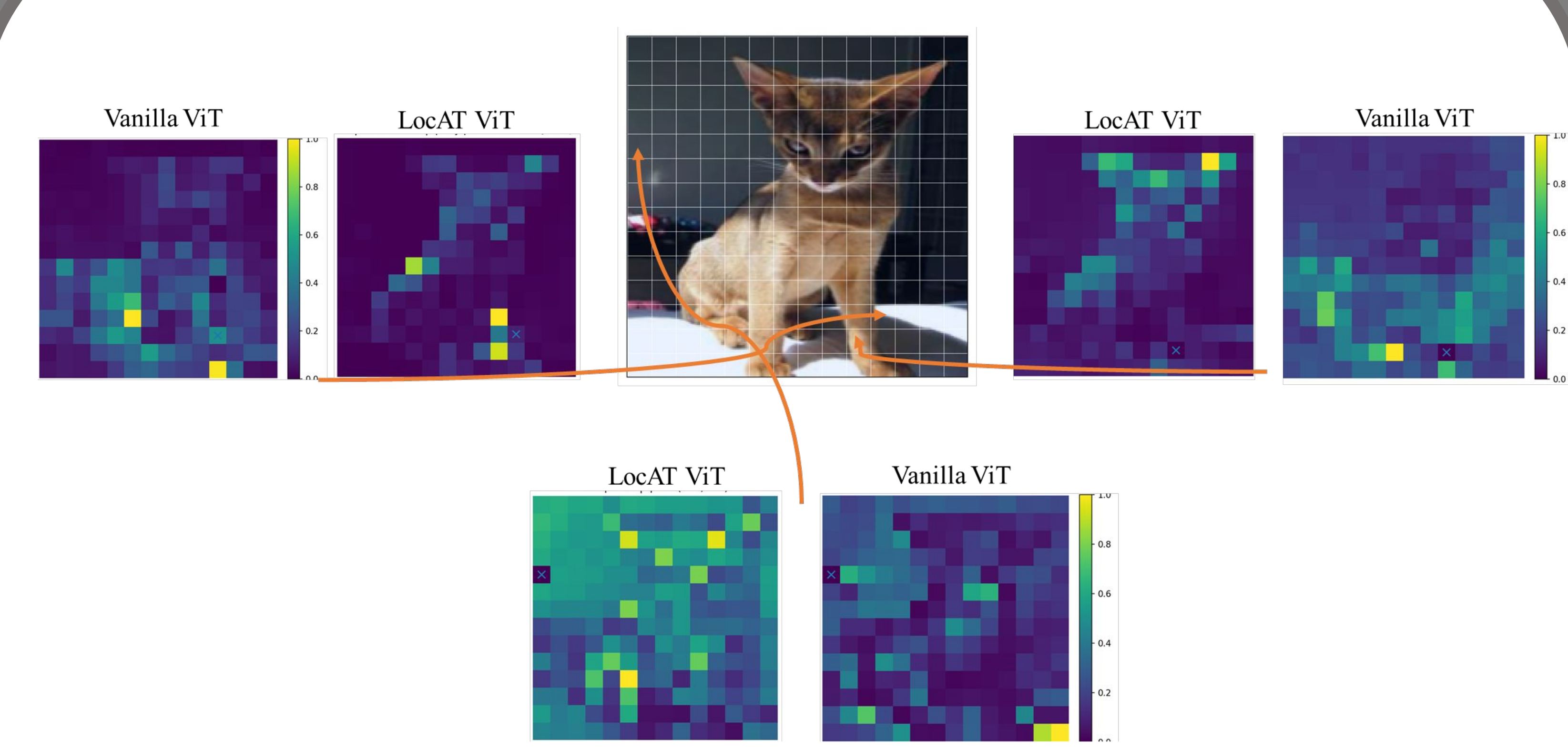
Metrics



Model	CIFAR-10 Test Accuracy (%)	Oxford-IIIT Pet mIoU (%)
Vanilla ViT	77	63.8
LocAT-ViT	**80.8**	**66.4**

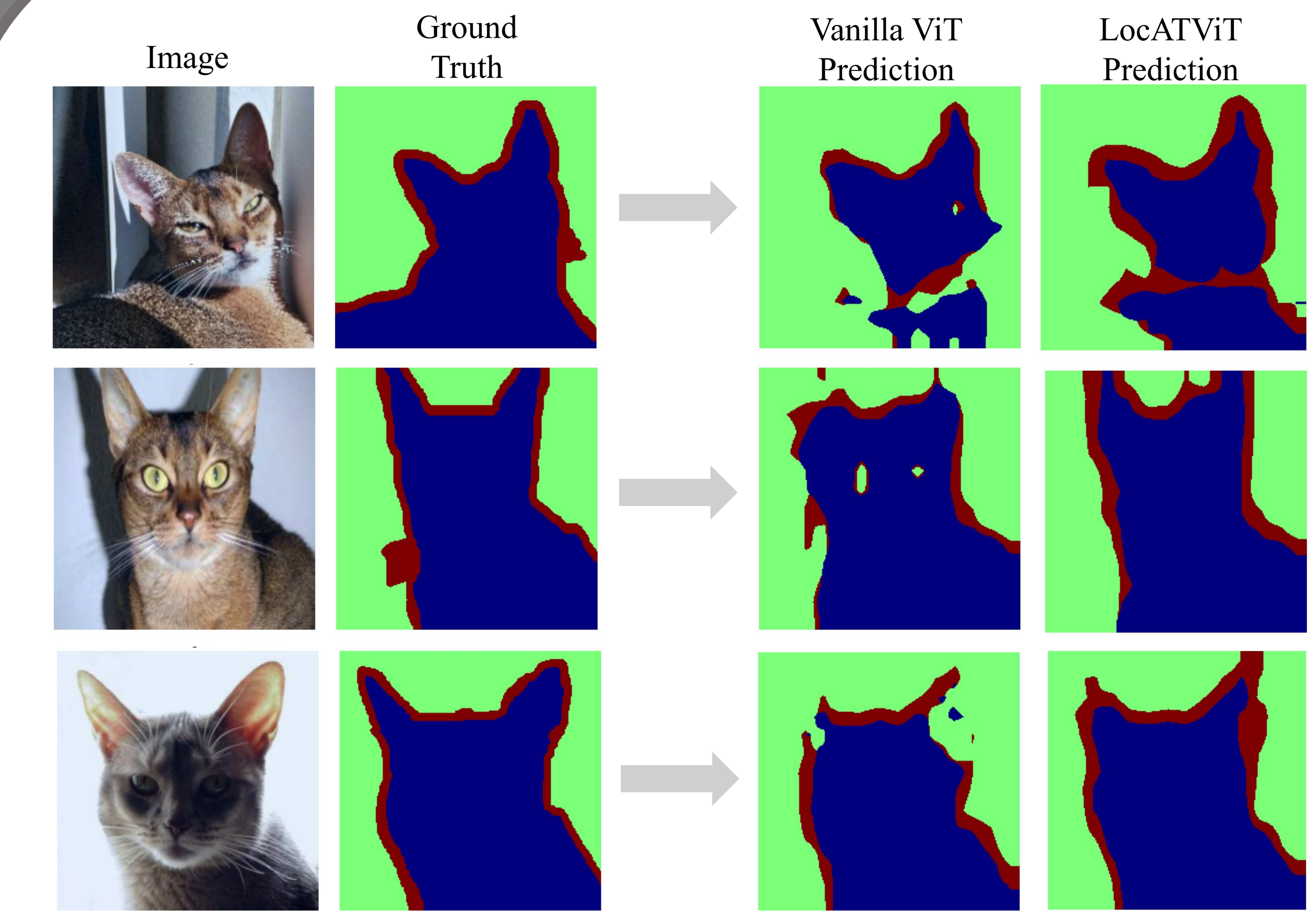
Table 1: Classification and Segmentation comparison between Vanilla ViT and LocAT-ViT.

Results: Attention maps



The final attention map of vanilla ViT and LocAtViT for three patches are illustrated for an image with label cat. LocAT introduces a spatial prior that favors nearby patches. This results in more spatially coherent attention maps, with attention concentrated around the query location.

Results: Segmentation



Qualitative segmentation results on Oxford-IIIT Pet. Compared to the vanilla ViT, LocAtViT produces smoother masks, fewer spurious predictions, and more accurate boundary alignment, indicating improved spatial feature learning. Green: background, Red: border, Blue: pet.

Conclusion Perspectives

Original Paper	Our Implementation
Relies on the `timm` library implementation of ViT	Implemented the architecture from scratch
Evaluated on large-scale datasets (ImageNet, ADE20K)	Used CIFAR-10 and Oxford-IIIT Pet due to computational constraints
Large-scale pretraining before downstream evaluation	Pretrained on smaller datasets
For segmentation, freezes the classification-trained backbone and trains a shallow 1-layer MLP head	freezes the classification-trained backbone and trains a lightweight 1x1 convolutional segmentation head

References

•Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale.* International Conference on Learning Representations (ICLR).